my files & folders

Documentation: Project Directory Structure

This document outlines the organization of the `/home/saurabh/moon_project/` directory. A consistent structure is essential for a reproducible scientific workflow.

Overview Tree

```
/home/saurabh/moon_project/
├── calib/
│   ├── CLASS/
│   └── XSM/
├── data/
│   ├── CLASS/
│   └── XSM/
├── maps/
├── metadata/
│   ├── metadata_CLASS/
│   ├── metadata_XSM/
│   ├── matched/
│   ├── filtered/
│   └── filtered_subsets/
├── plots/
├── results/
│   └── batch_run/
└── scripts/
```

`calib/`

- **Purpose:** To store all instrumental **calibration files**. These files are essential for converting raw instrument data into scientifically meaningful physical units.
- **Key Contents:**
    - **`calib/CLASS/`**: Contains calibration files for the CLASS instrument.
        - **`class_arf_v1.arf`**: The Ancillary Response File (ARF), describing the detector's sensitivity. Used by `py_forward_model_and_subtract.py`.
        - **`background_allevents.fits`**: The instrumental background template. Used by `py_forward_model_and_subtract.py`.
    - **`calib/XSM/`**: Contains calibration files for the XSM instrument.

        - `CH2xsmrspwitharea_open20191214v01.rsp`: The Response (RSP) file, containing both RMF (blur) and ARF (sensitivity) information.
        - `ch2_xsm_20200128_bkg.pha`: The instrumental background template for XSM.
        - Both files are used by the `pyxspec` analysis step inside `run_analysis_batch.py`.

`data/`

- **Purpose:** To store the original, raw scientific data downloaded from the ISRO PRADAN archive.
- **Key Contents:**
    - `data/CLASS/`: Contains the Level-1 calibrated `.fits` files from the CLASS instrument. Each file is a single lunar observation. These are the primary inputs for `perMonthEachDay_extract_class_metadata.py`.
    - `data/XSM/`: Contains the Level-1 calibrated `.fits` files from the XSM instrument. Each file contains a full day of per-second solar spectra. These are the primary inputs for `new_extract_xsm_metadata.py`.

`maps/`

- **Purpose:** To store the base map images used for final visualizations.
- **Key Contents:**
    - `lunar_map_16ppd.png` or similar: A high-resolution PNG of the Moon's surface (the LRO WAC global mosaic). This is used as the background image by `plot_abundance_map.py` and `plot_healpix_map.py`.

`metadata/`

- **Purpose:** To store all the intermediate CSV files generated during the data preparation and filtering phase (Phase 1). This separates the raw data from the processed catalogs.
- **Key Contents:**

- `metadata/metadata_CLASS/`: Contains daily CSVs of CLASS metadata, created by `perMonthEachDay_extract_class_metadata.py`.
- `metadata/metadata_XSM/`: Contains daily CSVs of XSM metadata, created by `new_extract_xsm_metadata.py`.
- `metadata/matched/`: Contains CSVs where CLASS and XSM data have been temporally matched, created by `match_CLASS_XSM.py`.
- `metadata/filtered/`: Contains CSVs of high-quality observations, created by `diagnose_and_filter_match_v1.2.py`.
- `metadata/filtered_subsets/`: Contains the final, premium list of observations to be analyzed (`match_CLASS_XSM_20250801_prime.csv`), created by `filteredFiles_subset_basedOnSolarAng.py`.

`plots/`

- **Purpose:** To store all visual outputs generated during the analysis.
- **Key Contents:**
  - **Diagnostic Plots (`.png`):** Images showing the quality of the scientific fits, such as the continuum subtraction and the final line fits, generated by `run_analysis_batch.py`.
  - **Final Maps (`.png`, `.html`):** The final 2D HEALPix maps and the interactive 3D `plotly` maps, generated by `plot_healpix_map.py` and `plot_3d_map.py`.

`results/`

- **Purpose:** To store the primary scientific outputs of the analysis pipeline.
- **Key Contents:**
  - `results/batch_run/`: This is the main working directory for the `run_analysis_batch.py` script.
    - **Intermediate Files:** For each observation, the script generates a set of temporary files: a co-added `.pha` file, an `XSPEC` command file `.cmd`, an unfolded spectrum file `_unfolded.txt`, and a residual spectrum file `_residual.txt`.

        - `final_abundance_ratios.csv`: The primary scientific output of the batch run. This file contains the measured line fluxes and abundance ratios for every successfully processed observation.
        - `final_abundance_ratios_complete.csv`: The final, enriched data file created by `merge_results.py`, which combines the scientific results with the footprint vertex coordinates. This is the main input for the final mapping scripts.

`scripts/`

- **Purpose:** To store all the Python source code (`.py` files) for the entire project.
- **Key Contents:**
    - **Phase 1 Scripts:** `perMonthEachDay...`, `new_extract_xsm...`, `match_CLASS_XSM...`, etc.
    - **Phase 2 Scripts (The Pipeline):**
        - `run_analysis_batch.py`: The main driver script that automates the entire scientific analysis.
        - `prepare_xsm_for_xspec.py`: Helper to create `.pha` files.
        - `py_calibrate_xsm.py`: Helper to process `XSPEC` output.
        - `py_forward_model_and_subtract.py`: Helper containing continuum subtraction logic.
        - `py_fit_xrf_lines.py`: Helper to perform the final line fitting.
    - **Phase 3 Scripts:** `merge_results.py`, `plot_healpix_map.py`, `plot_3d_map.py`.